

Machine Learning syllabus

Dr. Mohammed Brahimi and Dr. Aicha Boutorh



Outline

Machine learning Team

- Lecturers
 - Dr. Mohammed Brahimi
 - Dr. Aicha Boutorh
- Tutorial instructors
 - Dr. Mohammed Brahimi
 - Dr. Zineb Djouamai
 - Dr. Elhocine Boutellaa
- Lab instructors
 - Dr. Aicha Boutorh
 - Dr. Seif Eddine Bouzian
 - Dr. Youcef Omari
 - Dr. Ikram Remadna
 - Dr. Nasreddine Guelfout

Course Goals

- **Foundations:** Introduction to fundamental machine learning algorithms and their underlying principles.
- **Alternative Techniques:** Exploration of alternatives to the basic machine learning algorithms.
- **Algorithm Validation:** Equip students with robust validation techniques for machine learning algorithms to prevent erroneous conclusions.
- **Applied Learning:** Focus on the practical application and implementation of machine learning in real-world scenarios.

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Course Prerequisites

- **Mathematics:** Linear Algebra, Calculus, Probability, Statistics.
- **Programming:** Proficiency in Python/R or similar.
- **Data Handling:** Experience with data manipulation and analysis tools.
- **Algorithms:** Basics of algorithms, computational complexity.
- **Analytical Skills:** Critical thinking, logical problem-solving.

Course content

- **INTRODUCTION (1 lecture, 4%)**
 - Introduction to Machine Learning
- **PREDICTIVE DATA ANALYTICS (17 lectures, 71%)**
 - Information-Based Learning
 - Similarity-Based Learning
 - Probability-Based Learning
 - Error-Based Learning
 - Artificial Neural Networks (ANN)
 - Evaluation
- **Advanced Learning Techniques (4 lectures, 25%)**
 - Semi-supervised Learning
 - Active learning

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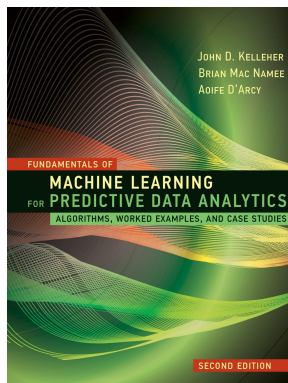
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Assessment method

- Final Exam â 60%
- Continuous Assessment â 40%
 - Midterm â 15%
 - Machine Learning Project â 15%
 - Quizzes â 5%
 - Attendance (lectures, tutorials, and labs) â 5%
 - A lecture attendance system will be implemented.

Textbooks

The main textbook: **Fundamentals of Machine Learning for Predictive Data Analytics.**



Other textbooks

These textbooks can serve as additional resources to deepen your comprehension of machine learning.

